



Rust Programming

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1 Rust Basics

1.1 Installing Rust

Install Rust from the [Rust website](#). Once you have rust installed, check that the Rust compiler is set up correctly by running

```
1 $ rustc --version
```

Rust can be updated using rustup, the included tool chain for installing and maintaining Rust. The version of rustup can be checked in a similar matter. Rust can be updated via rustup by running

```
1 $ rustup --version
2 $ rustup update
```

1.2 Cargo Basics

Rust runs typically from a main function created by initializing the project using cargo. Initialize a project using cargo

```
1 $ cargo new project_name
```

If the directory already exists, you can build a project in that directory using

```
1 $ cargo init
```

You can add dependencies, like external libraries from crates.io, by running

```
1 $ cargo add package_name
```

Finally, when you're ready, you can build your project for debugging or release using cargo as well via

```
1 $ cargo build # compiles for debug by default
2 $ cargo build --release # for building using the release profile
3 $ cargo run # compiles for debug and runs the executable
4 $ cargo run --release # compiles and runs the release profile executable
```

Additionally, you can always compile your Rust script using the Rust compiler itself and then running the produced executable

```
1 $ rustc main.rs # compile
2 $ ./program_name # run the produced executable
```

1.3 Basic Setup

Typical Rust programs will contain the following code snippet within `main.rs` by default

```
1 fn main(){
2     println!("Hello, world!")
3 }
```

The code contained in the main function is what gets compiled and called by the typical Rust compilation actions. Note that the default function setup contains the macro `println!` which prints the string literal `"Hello, world!"` to the console. We know this is a macro because the standard naming scheme for macros in Rust is to leave an `!` at the end of a function name.

1.4 Including Libraries

Certain libraries are packaged with Rust, known as the *standard library*, shortened to *std*. These packages do not need to be included as dependencies via cargo and can be brought into your code via the following syntax

```
1 use std::io
```

The double colon `::` is similar to the dot `.` notation from other languages where the above brings in the `io` library from the standard library.

1.5 Variable Declarations

New variables are declared using `let`. Variables are *immutable* by default and must be declared as *mutable* using the keyword `mut` during the variable declaration

```
1 let x = 5; // x is an immutable variable
2 let mut y = 6; // y is a mutable variable
```

You can, and should, annotate the type of a variable when declaring it like this

```
1 let x: i32 = 297;
2 let y: &str = "hello"
```

Note that `&str` is a reference to the string type which will be touched on later.

1.6 User Input

Input from the terminal can be obtained using the standard library's `io::stdin()` method.

```
1 io::stdin()
2     .read_line(&mut guess);
```

Where a mutable reference to the variable "guess" is passed to the standard io method allowing it to pass the terminal input to the location in memory where the variable guess is stored. This reference must be mutable because we are changing the value stored at that location. We can add the `except` method to this call and add a response in the case that the method fails.

```
1 io::stdin()
2     .read_line(&mut guess)
3     .except("Failed to read input");
```

2 Variables and Typing

Constants are declared with the keyword `const` and like most variables are immutable. The difference is that constants can *never* be made mutable.

```
1  const varname = 5;
```

Variables can also be reassigned via *shadowing* where an immutable variable has its value reassigned by re-declaring it using the current value with some modification added

```
1  let x = 5 // x is an immutable variable
2  let x += 5 // x now equals 10! We have changed an immutable variable
```

Note that this allows us to both change a variables value **and** its type because we are effectively creating a new variable with the same name.

3 Control Flow

Control flow is a basic and essential part of code which allows for controlling the path a written section of code takes based on designated inputs and information. Looping and conditional execution are two common examples of control flow.

3.1 Loop

Basic looping can be accomplished with the **loop** keyword which establishes a loop that continues until **break** is run

```
1  loop 'loopname {
2      println!("Looping!");
3      break 'loopname
4  }
```

Note that loops can also be named and continued or broken by their name. This is especially useful for controlling nested loops.

3.2 If Expressions

3.3 Scalar Data Types

Scalar data types contain data of only one type and fall into one of the following four categories

- Integers
- Floating Point Numbers
- Booleans
- Chars

Table 1: Integer Types in Rust

Length	Signed	Unsigned	Size of Values
8-bit	i8	u8	$2^8 = 256$
16-bit	i16	u16	$2^{16} = 65,536$
32-bit	i32	u32	$2^{32} \cong 4.294 \times 10^9$
64-bit	i64	u64	$2^{64} \cong 18.446 \times 10^{18}$
128-bit	i128	u128	$2^{128} \cong 3.40 \times 10^{38}$
Architecture-dependent	isize	usize	See above

3.3.1 Integers

Integers are whole numbers without decimal components and can be defined based on their size. Numeric literals that can be interpreted as multiple types can be specified by annotating them within their use.

```
1 let x: i32 = 23 + 16u16
```

Numeric literals can be specified in any of the following formats. If you're unsure

Table 2: Integer Literals in Rust

Number Literal	Example
Decimal	98_222
Hex	0xff
Octal	0o77
Binary	0b1111_0000
Byte (u8 only)	b'A'

what to use, Rust's default integer type is `i32`.

3.3.2 Floating Points

Floating point numbers can be specified as `f32` or `f64` and are represented according to IEEE-754.

3.3.3 Booleans

Booleans are values that can either be true or false, and are annotated as `bool`. Note that Rust, unlike some other languages, will not interpret any non-boolean type values as a `bool`. If the condition of non-boolean type data is checked, it will fail.

3.3.4 Chars

Chars are a 4-byte type representing a Unicode scalar value.

3.4 Compound Types

Compound types are data types which themselves may contain multiple different pieces of data which may be different types of their own

3.4.1 Tuples

Tuples are types of fixed length which may contain multiple different data types. Note that once a tuple is declared, its length may not be increased or decreased, though it may be de-structured into individual pieces

```
1 // declaring a tuple of different types
2 let tup: (i32, f64, u8) = (500, 6.4, 1);
3
4 // de-structuring a tuple
5 let (x, y, z) = tup;
```

You may also access individual elements of a tuple using dot notation

```
1      // declaring a tuple of different types
2      let tup: (i32, f64, u8) = (500, 6.4, 1);
3
4      // retrieving the first value from a tuple
5      let x = tup.0;
```

Empty tuples with no type have their own name and type, *unit*, written as `()`. Any expression which does not inherently return a value will return *unit*.

3.4.2 Arrays

Arrays hold collections of multiple items which have the *same* type. Like tuples, arrays have a fixed length. Arrays are declared with brackets and only need one type declaration along with their length.

```
1  // declaring an array of type i32 and length 5
2  let a: [i32; 5] = [1, 2, 3, 4, 5];
```

Array values are accessed using bracket notation

```
1  let first = a[0];
2  let second = a[1];
```

4 Functions

Functions are the main way of executing code in Rust and are declared with the `fn` keyword. The usual entry point of a Rust program is the `main` function that is created with a new project by default

```
1  fn main() {
2
3  }
4
5  fn secondary(x: i32) -> i32 {
6      x + 23
7  }
```

Functions can take inputs specified during the function signature of the annotated type and return a value of the specified type at the end of the function signature. Note that above, the value $x + 23$ is returned because it is the last expression in the function. A value elsewhere in the function can be returned using the keyword `return` followed by an expression.

This brings up the concept of expressions versus statements. The difference between the two is that expressions return a value while statements do not. Statements are terminated with a semicolon, while expressions are not.

